Experiment No 3

```
#include<stdio.h>
 #include<conio.h>
 #include<string.h>
 void main()
 {
   char a[10],ad[10],label[10],opcode[10],operand[10],symbol[10],ch;  int
st,diff,i,address,add,len,actual_len,finaddr,prevaddr,j=0;
   char mnemonic[15][15]={"LDA","STA","LDCH","STCH"};
   char code[15][15]={"33","44","53","57"};
   FILE *fp1,*fp2,*fp3,*fp4;
   clrscr();
   fp1=fopen("ASSMLIST.DAT","w");
   fp2=fopen("SYMTAB.DAT","r");
   fp3=fopen("INTERMED.DAT","r");
   fp4=fopen("OBJCODE.DAT","w");
   fscanf(fp3,"%s%s%s",label,opcode,operand);

   while(strcmp(opcode,"END")!=0)
   {
    prevaddr=address;
    fscanf(fp3,"%d%s%s%s",&address,label,opcode,operand);
   }
   finaddr=address;
   fclose(fp3);
   fp3=fopen("INTERMED.DAT","r");

   fscanf(fp3,"%s%s%s",label,opcode,operand);
   if(strcmp(opcode,"START")==0)
   {
    fprintf(fp1,"\t%s\t%s\t%s\n",label,opcode,operand);
    fprintf(fp4,"H^%s^00%s^00%d\n",label,operand,finaddr);
    fscanf(fp3,"%d%s%s%s",&address,label,opcode,operand);
    st=address;
    diff=prevaddr-st;
    fprintf(fp4,"T^00%d^%d",address,diff);
   }
   while(strcmp(opcode,"END")!=0)
   {
    if(strcmp(opcode,"BYTE")==0)
    {
     fprintf(fp1,"%d\t%s\t%s\t%s\t",address,label,opcode,operand);
     len=strlen(operand);
     actual_len=len-3;
```

```
        fprintf(fp4,"^");
 for(i=2;i<(actual_len+2);i++)
        {
  itoa(operand[i],ad,16);
  fprintf(fp1,"%s",ad);
  fprintf(fp4,"%s",ad);
        }
 fprintf(fp1,"\n");
}
else if(strcmp(opcode,"WORD")==0)
{
 len=strlen(operand);
 itoa(atoi(operand),a,10);
 fprintf(fp1,"%d\t%s\t%s\t%s\t00000%s\n",address,label,opcode,operand,a);
 fprintf(fp4,"^00000%s",a);
}
else if((strcmp(opcode,"RESB")==0)||(strcmp(opcode,"RESW")==0))
 fprintf(fp1,"%d\t%s\t%s\t%s\n",address,label,opcode,operand);
else
{
 while(strcmp(opcode,mnemonic[j])!=0)
  j++;
 if(strcmp(operand,"COPY")==0)
  fprintf(fp1,"%d\t%s\t%s\t%s\t%s0000\n",address,label,opcode,operand,code[j]);
        else
        {
  rewind(fp2);
  fscanf(fp2,"%s%d",symbol,&add);
   while(strcmp(operand,symbol)!=0)
    fscanf(fp2,"%s%d",symbol,&add);
  fprintf(fp1,"%d\t%s\t%s\t%s\t%s%d\n",address,label,opcode,operand,code[j],add);
  fprintf(fp4,"^%s%d",code[j],add);
        }
 }
 fscanf(fp3,"%d%s%s%s",&address,label,opcode,operand);
}
fprintf(fp1,"%d\t%s\t%s\t%s\n",address,label,opcode,operand);
fprintf(fp4,"\nE^00%d",st);
printf("\n Intermediate file is converted into object code");
fcloseall();

printf("\n\nThe contents of Intermediate file:\n\n\t");
fp3=fopen("INTERMED.DAT","r");
ch=fgetc(fp3);
while(ch!=EOF)
{
```

```
 printf("%c",ch);
 ch=fgetc(fp3);
} printf("\n\nThe contents of Symbol Table :\n\n");
fp2=fopen("SYMTAB.DAT","r");
ch=fgetc(fp2);
while(ch!=EOF)
{
 printf("%c",ch);
 ch=fgetc(fp2);
}
printf("\n\nThe contents of Output file :\n\n");
fp1=fopen("ASSMLIST.DAT","r");
ch=fgetc(fp1);
while(ch!=EOF)
{
 printf("%c",ch);
 ch=fgetc(fp1);
}
printf("\n\nThe contents of Object code file :\n\n");
fp4=fopen("OBJCODE.DAT","r");
ch=fgetc(fp4);
while(ch!=EOF)
{
 printf("%c",ch);
 ch=fgetc(fp4);
}
fcloseall();
getch();
}
```

INPUT FILES:

INTERMED.DAT

intermed.dat | symtab.dat

File Edit View

```
      COPY    START   2000
2000  **      LDA     FIVE
2003  **      STA     ALPHA
2006  **      LDCH    CHARZ
2009  **      STCH    C1
2012  ALPHA   RESW    1
2015  FIVE    WORD    5
2018  CHARZ   BYTE    C'EOF'
2019  C1      RESB    1
2020  **      END     **
```

SYMTAB.DAT

intermed.dat | symtab.dat

File Edit View

```
ALPHA    2012
FIVE     2015
CHARZ    2018
C1       2019
```

Output:-

```
ALPHA    2012
FIVE     2015
CHARZ    2018
C1       2019


The contents of Output file :

        COPY    START   2000
2000    **      LDA     FIVE    332015
2003    **      STA     ALPHA   442012
2006    **      LDCH    CHARZ   532018
2009    **      STCH    C1      572019
2012    ALPHA   RESW    1
2015    FIVE    WORD    5       000005
2018    CHARZ   BYTE    C'EOF'  454f46
2019    C1      RESB    1
2020    **      END     **


The contents of Object code file :

H^COPY^002000^002020
T^002000^19^332015^442012^532018^572019^000005^454f46
E^002000_
```

OUTPUT FILES:

ASSMLIST.DAT

ASSMLIST.DAT

File Edit View

```
        COPY    START   2000
2000    **      LDA     FIVE    332015
2003    **      STA     ALPHA   442012
2006    **      LDCH    CHARZ   532018
2009    **      STCH    C1      572019
2012    ALPHA   RESW    1
2015    FIVE    WORD    5       000005
2018    CHARZ   BYTE    C'EOF'  454f46
2019    C1      RESB    1
2020    **      END     **
```

OBJCODE.DAT

OBJCODE.DAT

File Edit View

```
H^COPY^002000^002020
T^002000^19^332015^442012^532018^572019^000005^454f46
E^002000
```